

Chapter 6 Pulse Modulation

Objectives

- To explain sampling theorem
- To analyze sampling process
- To study pulse amplitude modulation (PAM)
- To illustrate time division multiplexing (TDM) method
- To study pulse code modulation (PCM)
- To describe quantization process
- To determine quantization noise
- To describe encoding process
- To determine transmission bandwidth
- To study differential pulse code modulation (DPCM)
- To study delta modulation (DM)
- To compare PCM and DPCM, PCM and DM systems

Reference: Chapter 7, A. V. Oppenheim and A. S. Willsky, *Signals and Systems*, 2nd edition, Prentice Hall, 1997.
Chapter 6, B.P. Lathi, *Modern Digital and Analog Communication Systems*, 4th edition, Oxford, 2009.
Chapter 3, S. Haykin, *Communication Systems*, 4th edition, John Wiley and Sons, 2001.

In analog modulation (or *continuous wave modulation*), the modulation is applied continuously in response to the analog information signal. In pulse modulation, the sinusoidal carrier is not continuous but rather a pulse train, and the modulation is applied in response to the pulse information signal.

The *sampling theorem* makes the transition of a continuous wave to a pulse train.

6.1 Introduction

A continuous-time signal can be completely represented and recoverable by its pulse train from knowing its values at equally spaced points if certain conditions are met. This important and useful property is known as the sampling theorem. For example, it is used in moving pictures, which is a sequence of discrete frames which are taken from a view of a continuously changing scene. When these frames are viewed fast enough, it would seem that we are watching the original continuous moving scene. Another example would be that of a bitmap picture, which is a grid of points, with each point corresponding to a sample of a spatially continuous scene in that picture. If the points are sufficiently close together, we get the sense that the picture itself would be spatially continuous.

The sampling theorem is important partly because of its role as the bridge between continuous and discrete time signals. This chapter will explore how a continuous-time signal can be represented by a discrete-time signal by using the fact that a continuous-time signal can be recovered from a sequence of its samples under certain conditions. In many situations, it is more desirable to process discrete-time signals rather than continuous-time signals. This is mainly because of the development of digital technology, making discrete-time systems cheap, lightweight, reproducible, and programmable. Sample would therefore be attractive method of using discrete-time system technology to represent and implement continuous-time signals and process them. Sampling is used to convert discrete to continuous time signals, and continuous to discrete time signals.

We will later introduce and build the concept of sampling and the process of reconstructing a continuous-time signal from its samples. Now we'll explore the conditions of which a continuous-time signal can be exactly recovered from its samples, and see what happens when the conditions are not met. After, we'll study the processing of continuous-time signals that have been converted to discrete-time signals through sampling, then study the sampling of discrete-time signals and the related concepts of decimation and interpolation.

6.2 Representation of a Continuous-Time Signal by its Samples: the Sampling Theorem

We would think that in the absence of any additional conditions or information, that we would not be able to generate a unique continuous time signal from a sequence of equally spaced samples. In Figure 6.1, we have 3 signals that have the same value at every spaced of interval of T ; that is,

$$x_1(kT) = x_2(kT) = x_3(kT)$$

Therefore, there are an infinite number of signals that can generate a given set of samples. However, as we will observe, if a signal is band limited (Fourier transform is zero outside a band of frequencies), and that if the samples are taken close enough together relative to the highest frequency in the signal, then the samples specify the signal, and a continuous-time signal version of it can be reconstructed perfectly. This result is known as the sampling theorem, and is very important in the practical application of signal and system analysis methods.

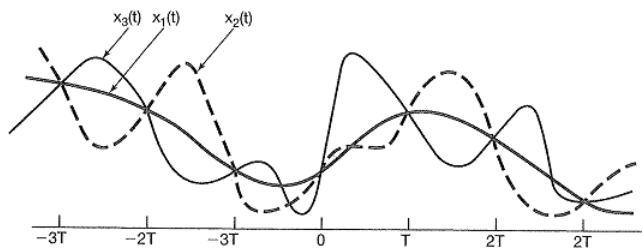


Figure 6.1 Three continuous-time signals with identical values at integer multiples of T .

6.2.1. Impulse-Train Sampling

We need to be able to mathematically express the sampling of continuous-time signals at regular intervals to develop the sampling theorem. A way to do this is to multiply a periodic impulse train by a continuous-time signal $x(t)$ which we wish to sample. Figure 6.2 graphically shows this method, also known as impulse-train sampling. The periodic impulse train $p(t)$ is known as the sampling function, T is the sample period, and the fundamental frequency of $p(t)$, $\omega_s = 2\pi/T$, as the sampling frequency. In the time domain,

$$x_p(t) = x(t) p(t) \quad (6.1)$$

where

$$p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT) \quad (6.2)$$

Because of the sampling property of the unit impulse discussed in Section 1.4.2, we know that multiplying $x(t)$ by a unit impulse samples the value of the signal at the point at which the impulse is located; i.e., $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$. Applying this to eq. (5.1), we see, as illustrated in Figure 6.2, that $x_p(t)$ is an impulse train with the amplitudes of

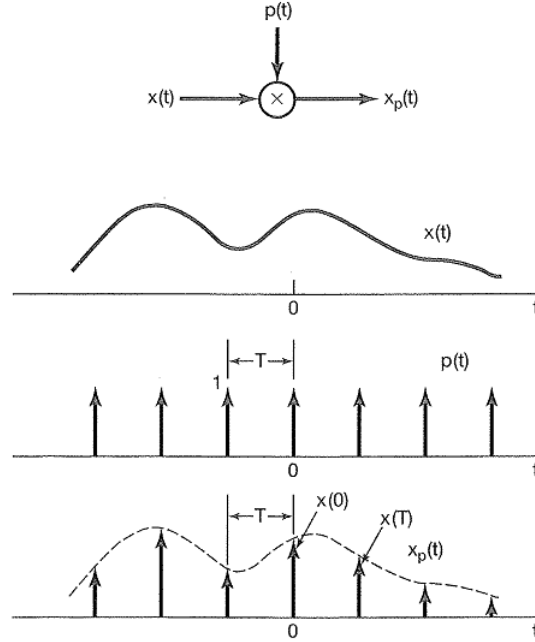


Figure 6.2 Impulse-train sampling.

the impulses equal to the samples of $x(t)$ at intervals spaced by T ; that is,

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x(nT) \delta(t - nT) \quad (6.3)$$

From the multiplication/convolution property of Fourier Transform, we know that

$$X_p(\omega) = \frac{1}{2\pi} X(\omega) * P(\omega) \quad (6.4)$$

and also

$$P(\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s) \quad (6.5)$$

Since $X(\omega) * \delta(\omega - \omega_0) = X(\omega - \omega_0)$ (Q12 of Hw 2), it follows that

$$X_p(\omega) = \frac{1}{T} \sum_{n=-\infty}^{+\infty} X(\omega - n\omega_s) \quad (6.6)$$

Figure 6.3 illustrates $X_p(\omega)$, which is a periodic function of time shifted $X(\omega)$ that is scaled by $1/T$. In Figure 6.3 (c), the spacing (or ω_s) between each shifted $X(\omega)$ is greater than $X(\omega)$'s width (or $2\omega_M$), so there is no overlap between each shifted $X(\omega)$. In Figure 6.3(d), ω_s is less than $2\omega_M$, so there is overlap. In Figure 6.3 (c), since there is no overlap, $X(\omega)$ can be reproduced at integer multiples of the sampling frequency. Similarly, if $\omega_s > 2\omega_M$, $x(t)$ can be 100% recovered from $x_p(t)$ by using a lowpass filter with gain T , and a cutoff frequency greater than ω_M and less than $\omega_s - \omega_M$, shown in Figure 6.4 The result of this is known as the *sampling theorem*.

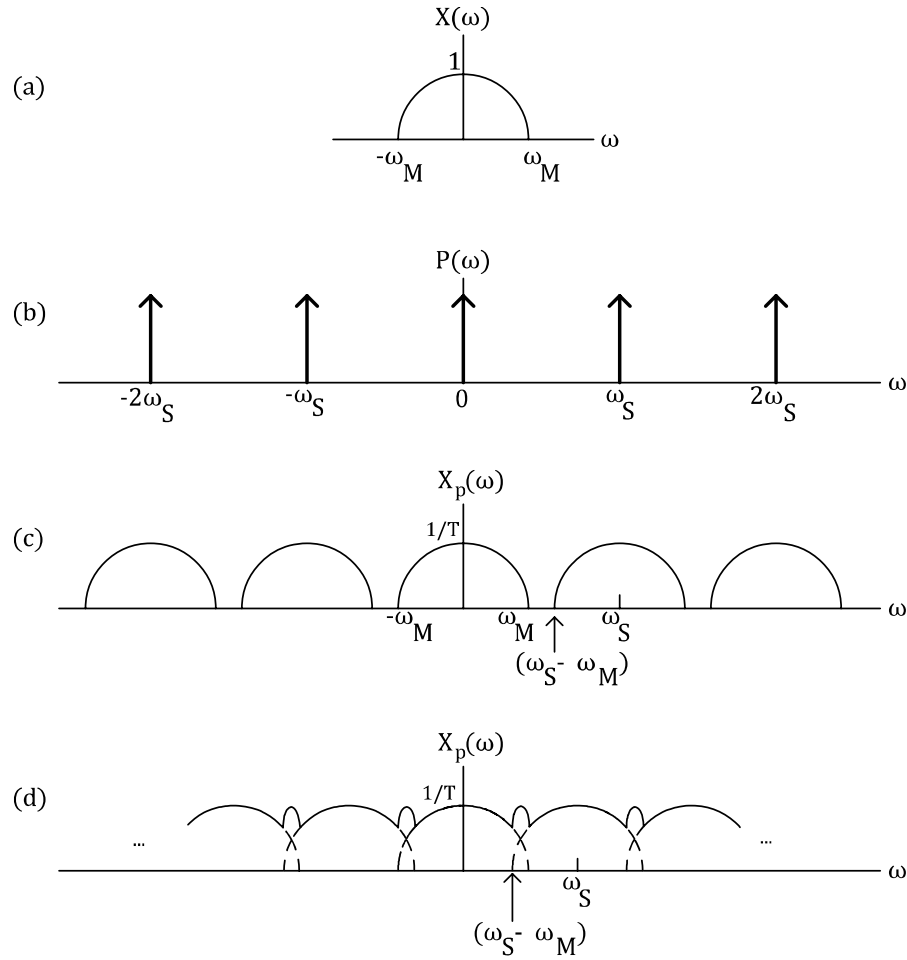


Figure 6.3. Effect in the frequency domain of sampling in the time domain: (a) spectrum of original signal; (b) spectrum of sampling function; (c) spectrum of sampled signal with $\omega_s > 2\omega_M$; (d) spectrum of sampled signal with $\omega_s < 2\omega_M$.

Sampling Theorem

Consider a band-limited signal $x(t)$ with $X(\omega) = 0$ for $|\omega| > \omega_M$. Let us then express $x(t)$ by picking sample points $x(nT)$, where n is an integer, and

$$\omega_s > 2\omega_M$$

where

$$\omega_s = 2\pi/T$$

We can now create a periodic impulse train where each successive impulse have amplitudes that are successive sample values. An ideal lowpass filter with gain T and cutoff frequency in between ω_M and $\omega_s - \omega_M$ can then be applied to the impulse train to exactly yield an output $x(t)$.

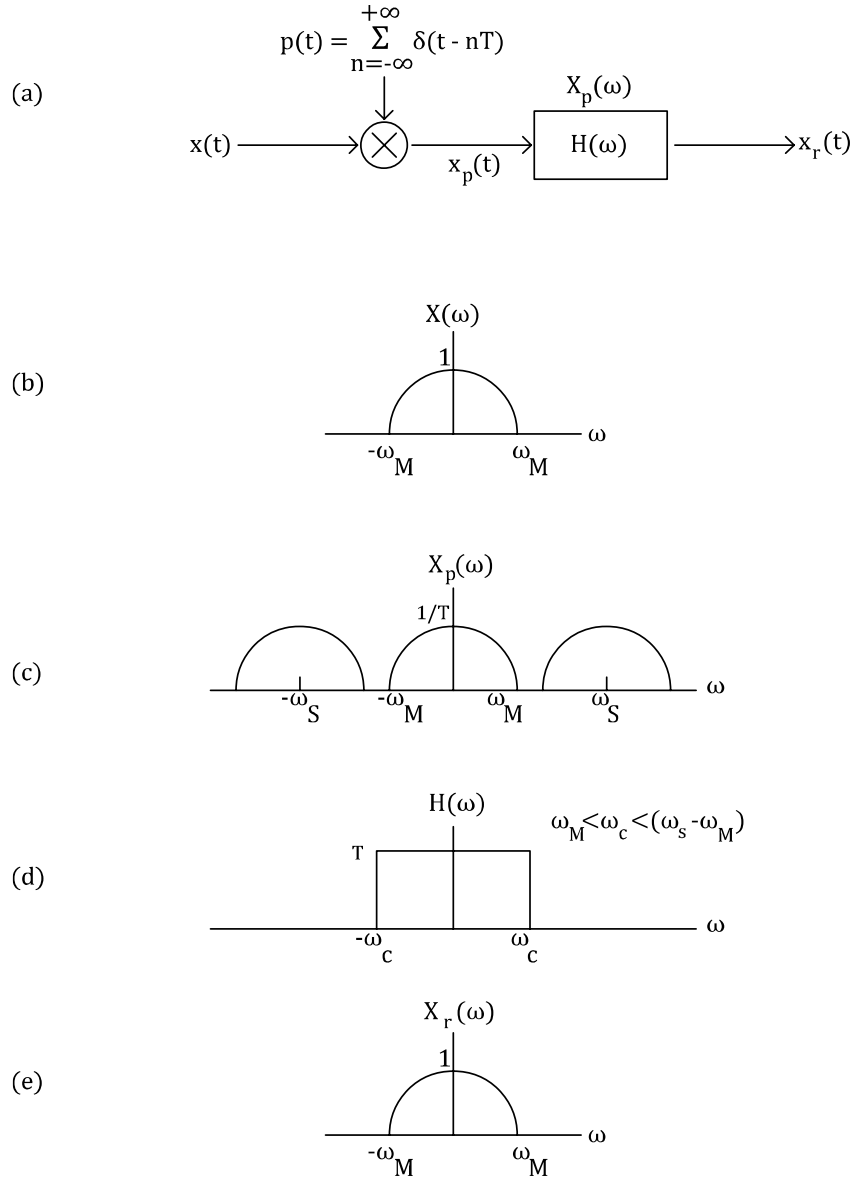


Figure 6.4. Exact recovery of a continuous-time signal from its samples using an ideal lowpass filter: (a) system for sampling and reconstruction; (b) representative spectrum for $x(t)$; (c) corresponding spectrum for $x_p(t)$; (d) ideal lowpass filter to recover $X(\omega)$ from $X_p(\omega)$; (e) spectrum of $x_r(t)$.

The sampling frequency must be greater than the frequency $2\omega_M$, and this is known as the Nyquist rate. Because ideal filters are not practical, non-ideal filters must be used instead and they are close enough to ideal for the problem of interest ($H(\omega) \approx 1$ for $|\omega| < \omega_M$, and $H(\omega) = 0$ for $|\omega| > \omega_S - \omega_M$). Since a non-ideal filter is used, the recovered signal $x_r(t)$ won't be exactly the same as $x(t)$ as shown in Figure 6.4. In this and the upcoming chapter, ideal filters will be used for convenience, although we must keep in mind that these filters are non-ideal in practice.

Example 6.1

Find the minimum sampling rate and the maximum sampling interval for the following signals.

a) $m(t) = 5\cos(1000\pi t)\cos(4000\pi t)$

b) $m(t) = \frac{\sin(200\pi t)}{\pi t}$

6.3 Reconstruction of a signal from its Samples Using Interpolation

Interpolation is a method in which a given number of sample points are used to reconstruct a signal either approximately, or exactly. Linear interpolation is one of the simplest forms of interpolation (shown in Figure 6.5) in which the sample points are connected to each other by a straight line. Other types of interpolation methods such as polynomial interpolation may use polynomials or mathematical functions to fit sets of sample points.

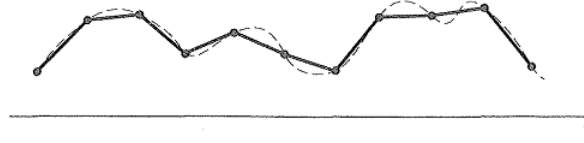


Figure 6.5 Linear interpolation between sample points. The dashed curve represents the original signal and the solid curve the linear interpolation.

Section 5.1 shows how a band-limited signal can be reconstructed exactly if samples are close enough together. In the time domain, exact interpolation can be used to reconstruct the signal exactly, which is actually the effect of the low-pass filter expressed in this domain. . In particular, the output is

$$x_r(t) = x_p(t) * h(t)$$

or, with $x_p(t)$ given by eq. (5.3),

$$x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT) h(t - nT) \quad (6.7)$$

Equation (5.9) is the interpolation formula of fitting a continuous curve between the sample points $x(nT)$. For the ideal low-pass filter $H(\omega)$ in Figure 6.4,

$$h(t) = \frac{\omega_c T \sin(\omega_c t)}{\pi \omega_c t} \quad (6.8)$$

so that

$$x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT) \frac{\omega_c T}{\pi} \frac{\sin(\omega_c (t - nT))}{\omega_c (t - nT)} \quad (6.9)$$

Figure 6.6 shows the reconstruction process of a band-limiting signal $x(t)$ using eq. (5.9) as the reconstruction method, with $\omega_c = \omega_s / 2$. Figure 6.6 (a) shows the original band-limiting signal $x(t)$, Figure 6.6 (b) shows the impulse train of samples, and Figure 6.6 (c) shows the superposition of the many terms in eq. (5.9).

This type of interpolation is commonly known as *band-limited interpolation*, since it exactly recovers $x(t)$ if the signal is band-limited, and if the sampling frequency is close enough to satisfy the sampling theorem. In practice, it might be more desirable to use a simpler albeit less accurate filter/interpolation function to recover a signal than in eq. (5.8).

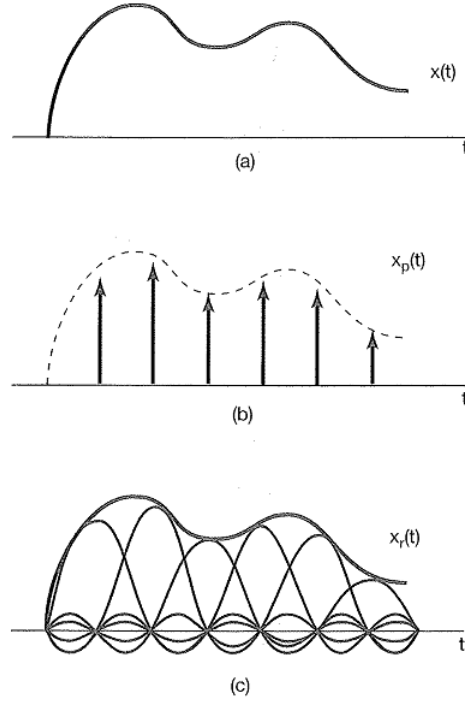


Figure 6.6 Ideal band-limited interpolation using the sinc function: (a) band-limited signal $x(t)$; (b) impulse train of samples of $x(t)$; (c) idea: band-limited interpolation in which the impulse train is replaced by a superposition of sinc functions [eq. (5.11)].

6.3.1. Signal reconstruction using Zero-Order Hold

The zero-order hold is a mathematical model to approximately reconstruct the input signal. It can be seen as an interpolation between sample values where the interpolating function $h(t)$ is the impulse response $h_0(t)$ shown in Figure 6.8. The function $h_0(t)$ is an approximation to the exact interpolation function of $h(t)$, similar to how $x_0(t)$ is an approximation $x(t)$. Figure 6.10 compares the magnitude of the zero-order hold filter to the ideal interpolating filter.

A problem arises because the zero-order hold does not have a constant gain in the passband. To mitigate this effect, we can process $x_0(t)$ generated from the zero-order hold with another LTI system with impulse response $h_r(t)$ and frequency response $H_r(\omega)$ shown in Figure 6.10. It would be ideal if we specify $H_r(\omega)$ such that the output $r(t) = x(t)$. We can get $r(t)$ to equal $x(t)$ if the combination of $h_0(t)$ and $h_r(t)$ yields the ideal low-pass filter $H(\omega)$. Since

$$H_0(\omega) = e^{-j\omega T/2} \left[\frac{2 \sin(\omega T / 2)}{\omega} \right], \quad (6.10)$$

the transfer function $H_r(\omega)$ will need to be

$$H_r(\omega) = \frac{e^{j\omega T/2} H(j\omega)}{2 \sin(\omega T / 2)} \quad (6.11)$$

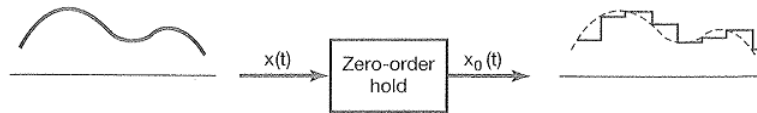


Figure 6.7 Sampling and reconstruction of signals utilizing a zero-order hold.

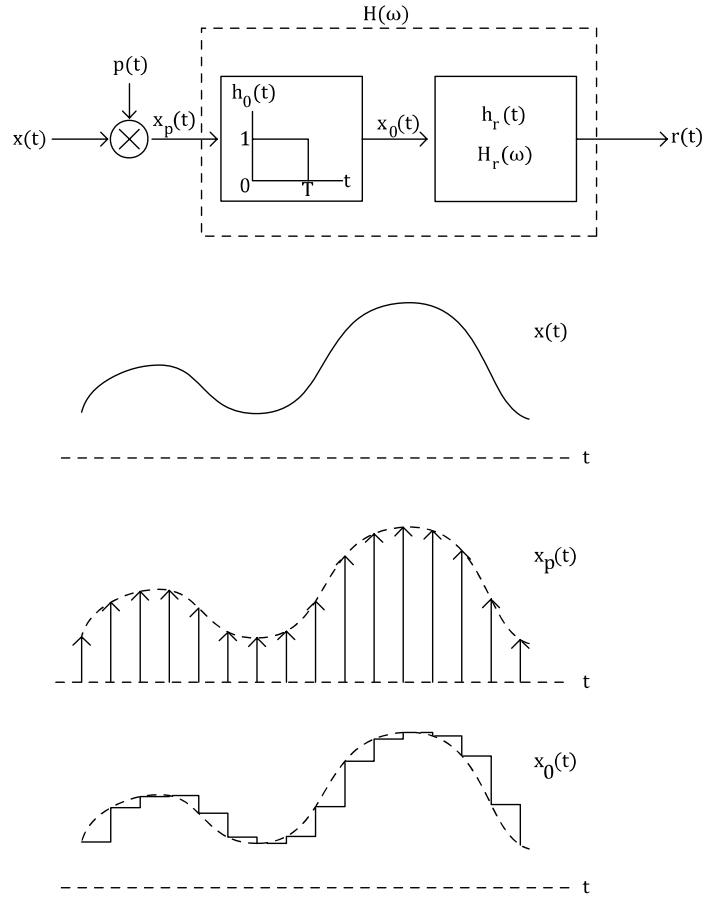


Figure 6.8 Zero-order hold as impulse-train sampling followed by an LTI system with a rectangular impulse response.

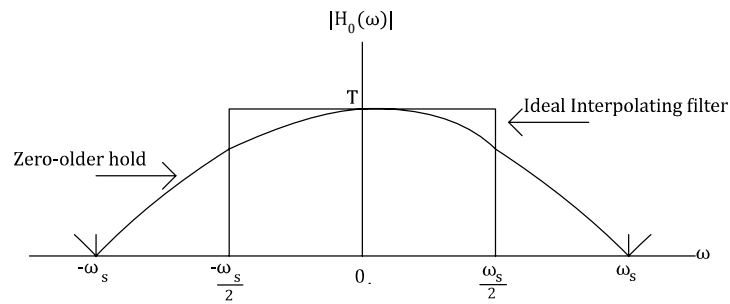


Figure 6.9 Transfer function for the zero-order hold and for the ideal interpolating filter.

Figure 6.10 shows the magnitude and phase for the reconstruction filter after a zero-order hold to reproduce $H(\omega)$ with a cutoff frequency of $\omega_s/2$. Equation 5.8 is hard to realize in practice, and must be approximated. The zero-order hold itself is actually an adequate approximation to the original signal without additional lowpass filtering. There are other situations in which we would need a better approximation between sample values.

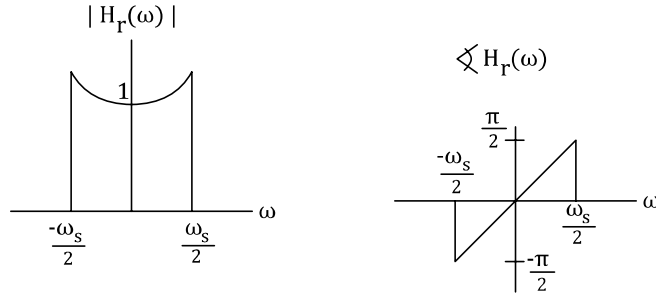


Figure 6.10 Magnitude and phase for the reconstruction filter for a zero-order hold.

6.3.2. Signal reconstruction using First-Order Hold

Previously, we said that the zero-order hold is adequate, although we can also see that it is very rough. We might desire a better interpolation/approximation, and these are called *higher-order holds*. The zero-order hold outputs a discontinuous signal. Linear interpolation, also known as a first-order hold, outputs a continuous signal, with discontinuous derivatives. Linear interpolation can be viewed as an interpolation similar to Figure 6.11 and equation 5.9, but the $h(t)$ is triangular. The transfer function for the latter is shown in the figure and is:

$$H(\omega) = \frac{1}{T} \left[\frac{\sin(\omega T / 2)}{\omega / 2} \right]^2$$

Figure 6.11 also compares the transfer function of the first-order hold to the ideal interpolating filter. We can use higher order holds that have higher degrees of smoothness. A second-order hold would be a continuous interpolation of the sample values, with a continuous first derivative, and discontinuous second derivative.

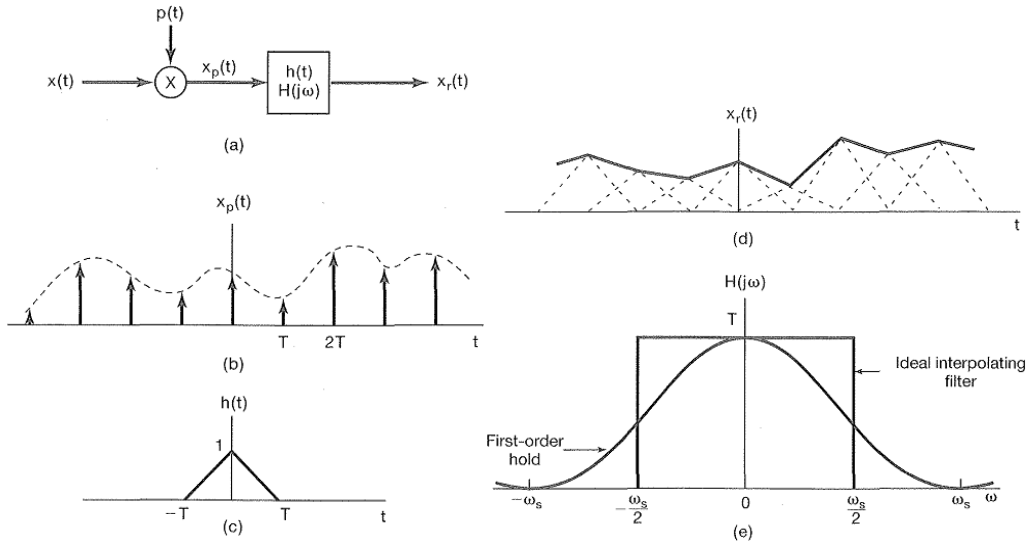


Figure 6.11 Linear interpolation (first-order hold) as impulse-train sampling followed by convolution with a triangular impulse response: (a) system for sampling and reconstruction; (b) impulse train of samples; (c) impulse response representing a first-order hold.

6.4 Practical Issues in Signal Sampling and Reconstruction

6.4.1. Realizability of Reconstruction Filters

Figure 6.12(a) shows a signal sampled at the Nyquist rate ($f_s = 2B$ Hz). The spectrum $\bar{G}(f)$ is a repetition of $G(f)$ without any gap between each of the cycles. Because there is no gap between each cycle, we would need $\bar{g}(t)$ to pass through an ideal-lowpass filter to recover $g(t)$ which is hard in practice to realize.

We can sample the signal at a higher rate ($f_s > 2B$ or $\omega_s > 4\pi B$) than the Nyquist rate to solve this problem. This yields $\bar{G}(f)$ consisted of a cycle of $G(f)$ with a gap in between each of them, shown in Figure 6.12 (b). $G(f)$ is now recoverable by using a lowpass filter with a gradual cutoff. However, the filter gain is required to be zero beyond the first cycle of $G(f)$.

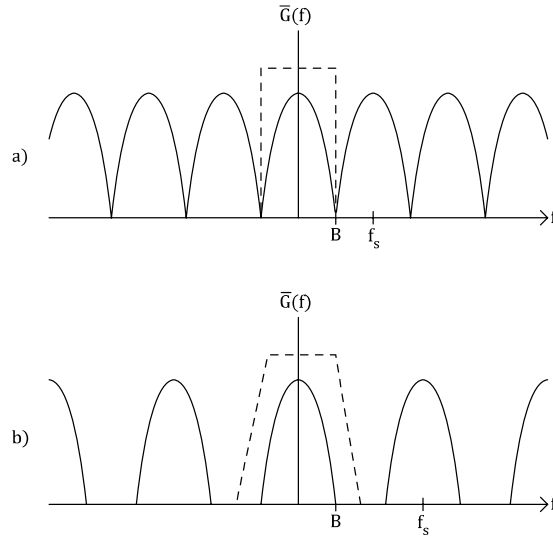


Figure 6.12. Spectra of a sampled signal: (a) at the Nyquist rate; (b) above the Nyquist rate.

6.4.2. Aliasing

Reconstructing a signal from its samples has another practical difficulty. The sampling theorem was based on the assumption that $g(t)$ was band-limited. In practice however, all signals are time-limited; they have a finite width. A time-limited signal cannot be band-limited, as well as the other way around. All practical signals which are time-limited, are therefore not band-limited, shown in Figure 6.13(a). The bandwidth for these signals are infinite, and $\bar{G}(f)$ consists of overlapping cycles of $G(f)$ repeated at the sampling frequency, shown in Figure 6.13 (b). Since the bandwidths are infinite, there will always be some spectral overlap between each repeated $G(f)$. Sampling at a higher rate will reduce but not completely eliminate spectral overlap. Therefore, $\bar{G}(f)$ does not have all the information of $G(f)$, and it would be impossible, be it theoretically or in practice, to recover $g(t)$ exactly from the sampled signal $g(t)$. Once the sample signal passes through an ideal low-pass filter of cutoff frequency $f_s / 2$ Hz, the output is a distorted version of $G(f)$, denoted by $G_a(f)$ in Figure 6.13 (c). The reason for this distortion is because of:

1. The loss of the tail of $G(f)$ beyond $|f| > f_s/2$ Hz.
2. The reappearance of this tail inverted or folded back onto the spectrum.

There is a spectrum cross at the frequency $f_s / 2 = 1/2T$ Hz, and this is known as the *folding frequency*. It is called this because it seems like the tail of the spectrum is folding back into itself at the folding frequency. This can be seen as the spectrum above the folding frequency for a repeated signal is the same shape (although horizontally reversed) as the spectrum below the folding frequency. Since this is overlapped with the next repeated signal, it gives the sense that the tail is folded back onto itself, and this is known as spectral folding/aliasing, shown shaded in Figure 6.13 (c). Therefore, in the aliasing process, not only are we losing the tail (frequencies above $f_s/2$), but they show up (aliased) in the lower frequency components, shown in Figure 6.13 (b), and (c). This will deform the propagated signal as a result.

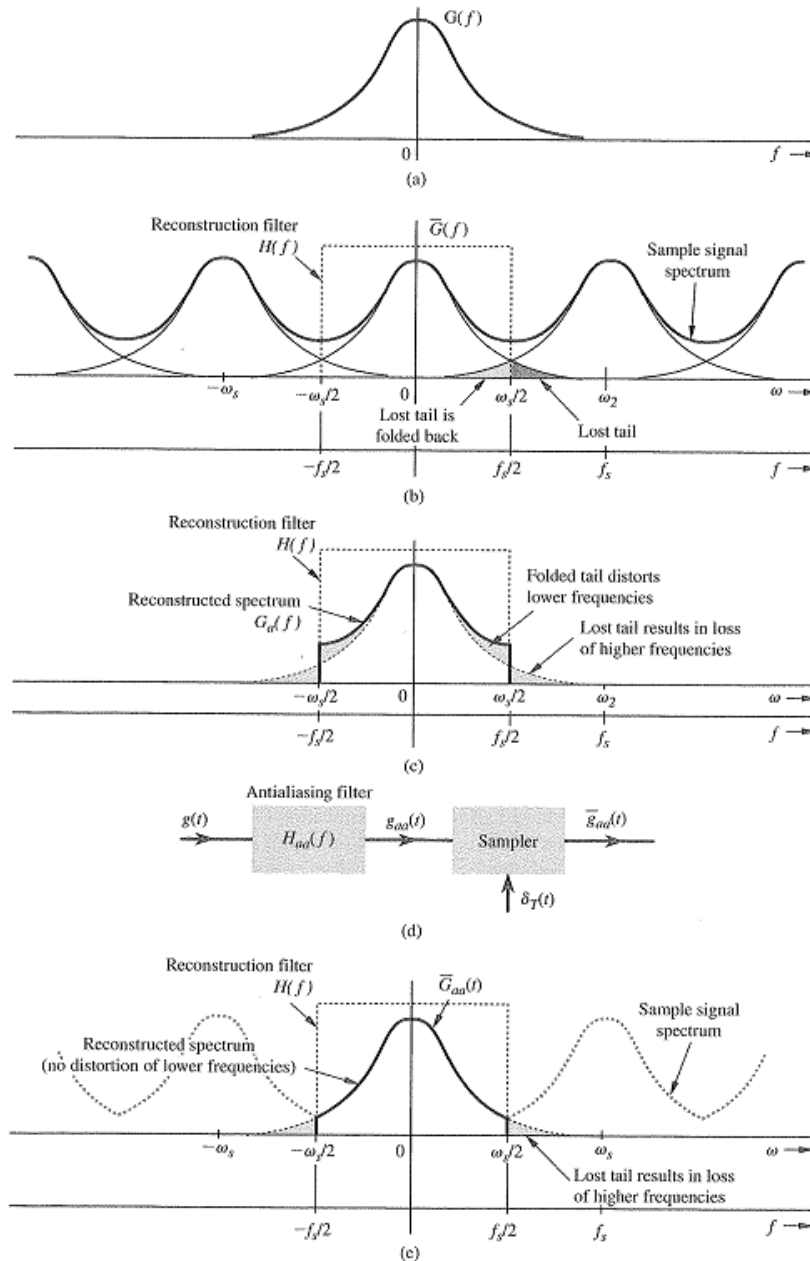


Figure 6.13 Aliasing effect. (a) Spectrum of a practical signal $g(t)$. (b) Spectrum of sampled $g(t)$. (c) Reconstructed signal spectrum. (d) Sampling scheme using antialiasing filter. (e) Sampled signal spectrum (dotted) and the reconstructed signal spectrum (solid) when antialiasing filter is used.

6.4.3. The anti-aliasing Filter

An anti-aliasing filter is a way to limit the effects of aliasing. It is done by eliminating all the frequency components of $g(t)$ that is greater than the folding frequency **before sampling it**. The elimination of these frequency components is done by applying an ideal low-pass filter with the cutoff at $f_s/2$ Hz before sampling. Figure 6.13 (e) shows a reconstructed signal $G_{aa}(f)$ when an anti-aliasing filter is used. By eliminating the frequency components that are greater than the folding frequency, these frequencies won't appear and corrupt the frequencies below the folding frequency. By using the anti-aliasing, the reconstructed signal $G_{aa}(f) = G(f)$ for $|f| < f_s/2$. Therefore, half of the aliasing distortion is cut in half due to folding. Of course, an ideal anti-aliasing filter is not practical, so steep-cutoff filter is used instead, which sharply attenuates the residual spectrum beyond the folding frequency.